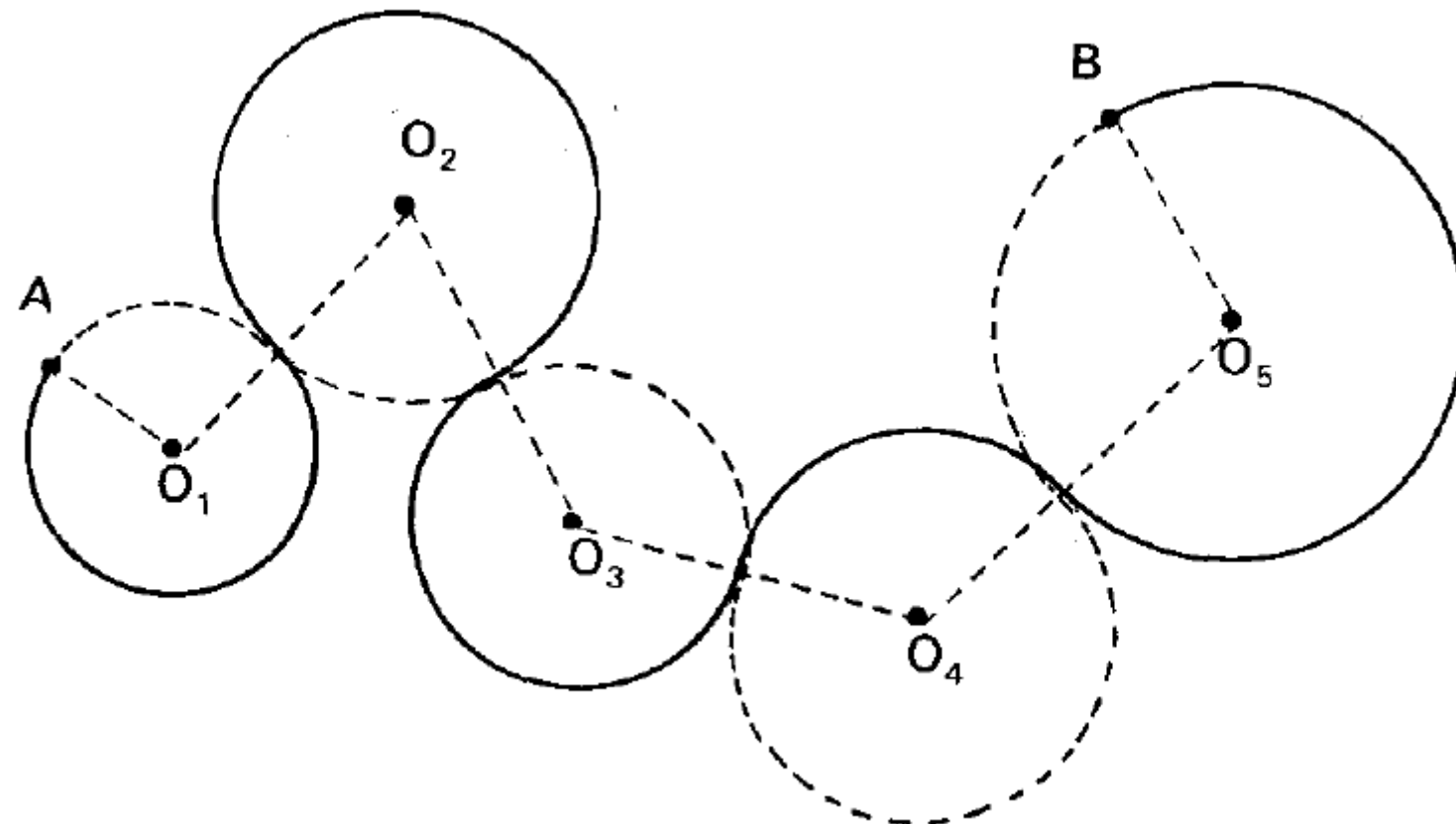


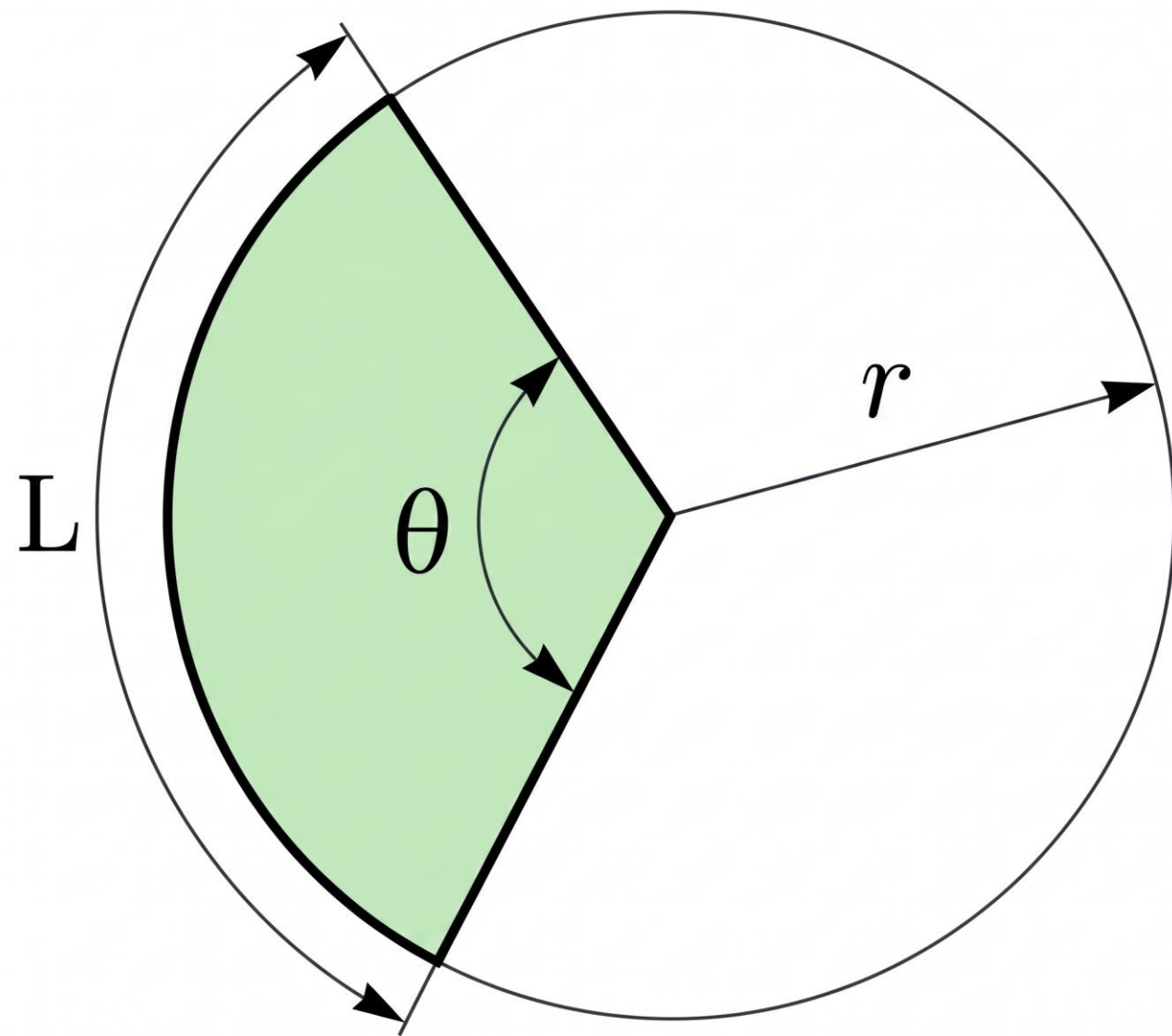
A fun geometry problem

If the angles at the vertices O_1, O_2, O_3, O_4, O_5 measure, respectively, $90^\circ, 72^\circ, 135^\circ, 120^\circ, 105^\circ$, and the radii of the circles centered at these vertices measure, respectively, 18 cm, 35 cm, 24 cm, 36 cm, 48 cm, determine the length of the line segment AB .



DOLCE, Osvaldo; POMPEO, José Nicolau. Fundamentos de Matemática Elementar: Geometria Plana. Vol. 9, 7ed. São Paulo: Atual Editora, 1993.
Question 737 at page 296.

A quick review



$$\frac{L}{\text{circumference}} = \frac{\theta}{2\pi}$$

$$\theta = \frac{\pi}{180} \alpha \quad (\text{Here } \theta \text{ is in radians and } \alpha \text{ in degrees})$$

$$L = \frac{\pi}{180} \alpha r$$

$$\sigma = 2\pi - \theta \quad (\text{explementary angles})$$

The solution

$$L_{AB} = \frac{\pi}{180} \left(\underbrace{\sum_{\{\theta_i\}}^m R_i \theta_i}_{\text{Inner Angles}} + \underbrace{\sum_{\{\sigma_i\}}^n R_i \sigma_i}_{\text{Explementary angles}} \right)$$

The solution

$$L_{AB} = \frac{\pi}{180} (18 \times 270^\circ + 35 \times 288^\circ + 24 \times 225^\circ + 36 \times 240^\circ + 48 \times 255^\circ)$$

$$L_{AB} = 205\pi$$

